

Pneumonia Detection System Using Fusion Datasets

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Abstract—Background: Typically, doctors diagnose pneumonia based on symptoms and chest X-rays. Now, using chest X-ray images and machine learning, new automated systems can be developed that reduce time and equipment and can be used in all places and situations where doctors and expensive equipment are unavailable. This pneumonia disease detection model will not only help in accurately identifying whether a person has pneumonia but will also save time. And it will also reduce the costs and other hassles that the person and his relatives face during the disease detection. We used a dataset of 10,448 chest X-ray images (7758 pneumonia cases and 2690 normal cases), which was divided into training, validation, and test sets. We also analyzed the dataset and generated various graphs. Next, we preprocessed the data to train the model. Our method uses a convolutional neural network (CNN) and validates it in terms of accuracy, precision, and recall. The proposed model achieves 38% accuracy, 39% recall, and 38% f1-score when detecting people without pneumonia. And achieves 63% accuracy, 62% recall, and 62% f1-score when detecting people with pneumonia. During validation, we obtain 88%, 93%, and 87% accuracy, recall, and recall, respectively. It can detect people with pneumonia well. In each scenario, we obtain very high precision compared to others. This system will be able to support doctors in making clinical decisions, which can significantly reduce the diagnosis time compared to conventional methods while maintaining high accuracy. This system will also benefit those areas where high-quality equipment is not available and installation is almost impossible. The results of the system will be further improved by considering the quality of the dataset, the effectiveness of pre-processing and the selection of appropriate datasets.

Index Terms—Chest X-ray, image type dataset, pneumonia, detection, machine learning, medical imaging, disease, CNN

I. INTRODUCTION

Pneumonia is a major respiratory disease that affects and kills a large number of people every year. Pneumonia is a severe disease affecting the lungs. When suffering from this, the lungs get filled with water, and there is swelling in them [1]. People of almost all ages are affected by this disease. However, children and the elderly face serious consequences of this disease. Community-dwellers with AD had a higher incidence of pneumonia than community-dwellers without AD [2]. It can be caused by bacteria, viruses, fungi. However, bacteria are the main cause of pneumonia. Increased levels of O_3 , SO_2 , and $PM_{2.5}$ are generally associated with increased pneumonia cases over a large area [3]. According to Yui Jeong Roh et al. [4], the causes of community-acquired pneumonia (CAP) vary depending on individual susceptibility, community epidemiological characteristics, and season. This disease disrupts the immune system of the human body.

The analysis of this research question is:

A pneumonia detection system that integrates fusion datasets such as clinical data, imaging data, and other risk factors and incorporates explainable artificial intelligence (XAI) methods such as SHAP, LIME, Grad-CAM, etc. to improve detection performance.

The remainder of this paper is organized as follows: Section [I] provides information about the paper, the model, research methods, and techniques for detecting pneumonia. Section [II] Review of past work on detection and discussion of limitations and solutions to overcome the limitations. The methodology is described in Section [III] Selecting datasets based on quantity and quality, then collecting and processing that data in an appropriate format. Data augmentation techniques were integrated to address imbalances in the dataset. The setup of the experiment is presented in Section [IV] Description and conclusion of the disease we are working on, and the method we are adopting to solve it. Technical descriptions such as programming languages, algorithms, hardware, and evaluation metrics for building the model. The result analysis is presented in Section [V] Developed CNN architecture for chest X-ray image classification. Diagnostic performance was evaluated using multiple metrics, including precision, accuracy, recall, F1-score, and AUC, and reliability was assessed extensively. Review the results we get after applying the dataset, the pre-processing step, running the model, and developing the model. Finally, Section [VI] Showing the entire work process, results, and areas that still need improvement. Model limitations were highlighted and compared with detailed error analysis, so it is easy to understand where the model is more effective.

II. RELATED WORK

Khalid et al. [7] presented the work with two publicly available image datasets using COVID-19 X-ray images. The first one consists of 5856 images of three classes, while the second one contains 231 COVID-19 chest X-ray images. The combined dataset consists of 6087 images in jpeg format. And they used three models: InceptionResNet_V2, ResNet50, MobileNet_V2. The evolution metrics used are accuracy, sensitivity, specificity, precision and F1-score. The accuracy and specificity of the InceptionResNet_V2 model are the highest compared to the other models.

Goyal et al. [8] used the C19RD dataset of chest X-rays. The model used 3 formats of datasets in Raw features, min-max normalization, and strong normalization. The strong normalization dataset outperformed other datasets. The models used were KNN, SVM, ANN, ENSEMBLE, F-RNN-LSTM. And the performance of F-RNN-LSTM model is much higher than other models. The accuracy of F-RNN-LSTM model is 95.04%, precision is 93.65%, recall is 96.78%, specificity is 94.21% and F1-score is 95.19%.

Karim et al. [9] used computer-aided techniques for pneumonia detection. Here, different pre-processing techniques such as X-ray are explored. The study examines how different machine learning techniques such as convolutional neural networks (CNN), k-nearest neighbors (KNN), RESNET, CheXNet, DECNET, and artificial neural networks (ANN) can be used in pneumonia disease detection. The proposed solution involves using a privacy-preserving approach that will allow the use of real-time data, which meets the requirement of having huge data and different types of medical images for ML model training.

Hedhoud et al. [10] proposed an explanation model based on Convolutional Neural Network (CNN) and Extreme Gradient Boosting (XGboost) for pneumonia classification. The experimental study is processed under different scenarios, using Python as a programming language and a public database obtained from the Guangzhou Women and Children's Medical Center. Then, 87% accuracy, 89% specificity, and 85% sensitivity were obtained. The dataset includes 5863 images of pediatric patients aged 1-5 years.

Age, gender, region also play an important role during pneumonia detection but most datasets lack data with such features. So training the model with only a few selected features and ignoring other features is a major limitation. There is also a lack of sufficient quality data for model training. In existing work, some models train with little data and the performance of that model can be improved by training with additional datasets. Model deployment is also an important step after training a model and getting a suitable model. In many cases, model deployment is also a major hurdle. In this technology, the trained model is involved in transporting it over the internet, and security can be compromised here. Also, adopting this technology will involve additional cost in terms of client resource consumption.

For our proposed model, we use a dataset that has a sufficient amount of data, and quality data exists in the dataset. In this way, the detection ratio can be greatly increased. Pre-processing of the quality dataset is one of the necessary steps. Pre-processing the dataset is a useful method to achieve the desired results and overcome other limitations faced during pneumonia disease detection. Appropriate model selection and selection according to model characteristics is also effective method. Another important issue for medical work is to achieve high accuracy because failure in disease detection can have very high consequences.

III. DATASET DESCRIPTION

The datasets are collected from Kaggle. The datasets consist of labeled chest X-ray images from Tolga Dincer [5] and chest X-ray images from Paul Mooney [6].

- **Sample Size:** Here we are using 10,448 images as sample images. 2,690 images are normal and 7,758 images have pneumonia.
- **Features:** All images resized to 150 image width $\times$ 150 image height
- **Class Distribution:** Unbalanced because only 25.7% are normal images and the remaining 74.3% are pneumonia images. From the pie chart in Figure 2, we can see that the difference between the two cases is 48.6%. The number of chest X-ray images of pneumonia is about 50% higher in the dataset. The bar chart also shows a similar difference.

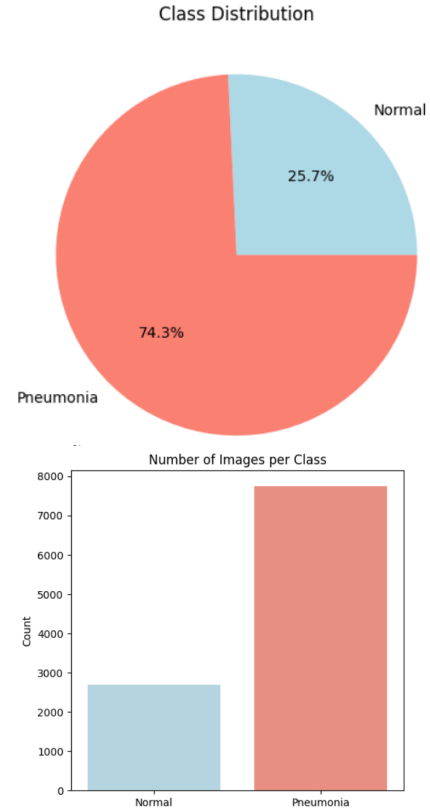


Fig. 1. Pie charts and bar charts for dataset analysis

All images were converted to 150×150 format. Batch size of 32 means that during training, the model processes 32 images at a time. The pixels of the X-ray images were normalized to the range $[0, 1]$ for neural network input. Data augmentation included random rotation by $\pm 20^\circ$ and random horizontal/vertical shifts of 20% of the image dimensions. The dataset was split into training (80%) and validation (20%) subsets.

Figure 2 shows sample chest X-ray images from the dataset we used. These images are manipulated and then used in the model to learn the prediction task.

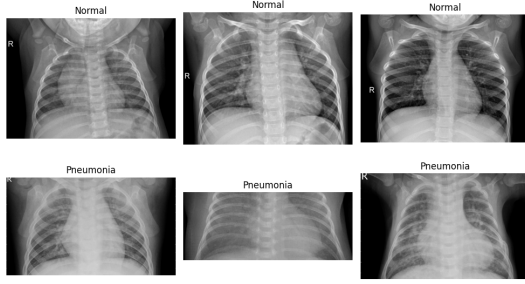


Fig. 2. Pneumonia and normal chest X-rays from the dataset

The imbalance between normal and pneumonia images in the dataset is very high and the model would perform better if it was trained by reducing the gap in the number of images. Here, no other information like age, gender, etc. was provided apart from the images and adding that information would also help. By increasing the quality and quantity of images, the model would improve its accuracy and performance.

IV. METHODOLOGY

The proposed model is a CNN designed to discriminate between chest X-ray images of normal people and pneumonia-affected people. The CNN learns from the image data through convolutional layers. Conv2D, Maxpooling2D, used to extract features from the images in the dataset, reduces dimensionality, handles overfitting, and reduces cost. Flatten() prepares the data for the Dense() layers by converting the 2D output to 1D vectors. Dropout() works during training. Dense() connects all the neurons and makes the final decision.

The log loss or binary cross-entropy loss function is widely used in real-world machine learning situations because it works effectively when measuring probabilities and the predicted probabilities are significantly higher in binary classification problems. It effectively estimates the correct probabilities. It is most needed in important areas like medical disease prediction. Because here high probability of prediction is mandatory.

The binary cross-entropy loss function is given by [11]:

$$\mathcal{L}(y, \hat{y}) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

- **Programming Language:** Python. (Used for simplicity and its library support.)
- **Framework:** TensorFlow (simplify model building, GPU and TPU support), Numpy (numeric computation), Pandas (data preprocessing), Matplotlib (pie charts, bar charts), Seaborn (heatmaps)

- **Libraries used:** Scalern (data analysis and modeling), OS (file operations), ImageDataGenerator (data augmentation)

Python, TensorFlow have been used as software. All Python code has been run on Google Colab and TensorFlow has also been run. The model has been run and built on Google Colab. And while building the model, computer CPU and Google Colab T4 GPU have been used.

Evaluation Metrics: The model's performance was evaluated accuracy, precision, recall, F1-score, and AUC.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- TP or True Positive means correctly identifying an infected person
- TN or True Negative means correctly identifying a normal person
- FP or False Positive means correctly identifying an infected person as a normal person
- FN or False Negative means correctly identifying a normal person as an infected person

V. RESULTS AND DISCUSSION

Convolutional Neural Network (CNN) achieved moderate classification performance on a dataset of chest X-rays for pneumonia diagnosis. The performance metrics show: accuracy 53%, precision 62.7%, recall 61.5%, and F1-score 62.1%. These metrics indicate the ability of the model to correctly select pneumonia and normal cases and most of the time the model identifies the appropriate cases. From the confusion matrix we see that the model is able to identify 96 pneumonia cases and 36 normal cases. In our study, we used 10,448 images of pneumonia and normal. Enes AYAN et al. [12] evaluated two models using 624 frontal chest X-ray images. The test set consisted of 234 normal and 390 pneumonia patients.

TABLE I
CNN MODEL CONFUSION METRICS

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.5301	0.6274	0.6153	0.6213

Figure 3 shows the confusion matrix, offering a breakdown of true/false positives and negatives, reinforcing the balanced performance across classes.

The table contains our classification performance metrics.

The dataset we provide contains more pneumonia images than normal. Here, support detects whether the dataset is unbalanced or not. Such unbalanced datasets significantly affect

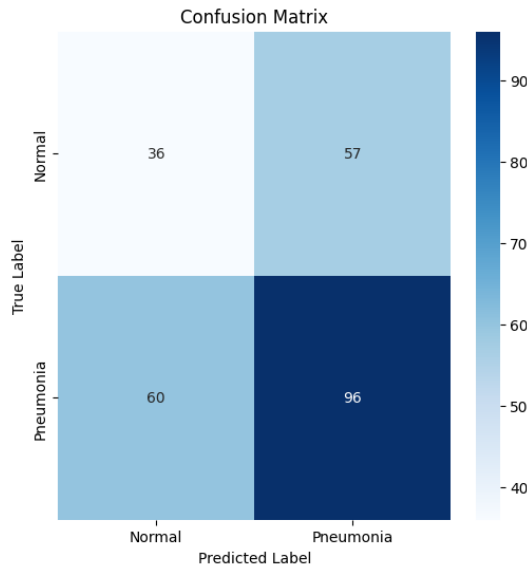


Fig. 3. Confusion Matrix of the CNN Model

the overall performance of the model because in that situation the model is more likely to over-focus on pneumonia. Support identifies 93 and 156 X-ray images as normal and pneumonia, respectively. Since the dataset is unbalanced, macro average and weighted average will work here. Macro average and weighted average play an effective role in evaluating the performance of the model without being affected by class imbalance. Here in both cases we get macro average of 0.50 and weighted average of 0.53. When the model predicts pneumonia, it correctly predicts 63 cases out of 100 cases and the remaining 37 cases are shown as normal. But only 38 normal cases are able to be identified. So the model is able to identify pneumonia patients much better than normal individuals.

From the bar chart in Figure 4, we can see that the precision, recall and F1-score for pneumonia detection are almost the same which is more than 60% and the accuracy is slightly lower which is less than 60%.

TABLE II
CLASSIFICATION PERFORMANCE METRICS

Class	Precision	Recall	F1-Score	Support
Normal	0.38	0.39	0.38	93
Pneumonia	0.63	0.62	0.62	156
Macro Avg	0.50	0.50	0.50	249
Weighted Avg	0.53	0.53	0.53	249

The imbalance between the number of pneumonia cases and the normal image in our dataset is significantly high. This is why the model performs well when predicting pneumonia cases but very poorly when predicting normal cases. This is why the number of false predictions increases. Here, the prediction ratio for pneumonia is more than 60%, but the prediction ratio for normal cases is less than 40%. In the medical field, an accuracy of more than 90% is almost

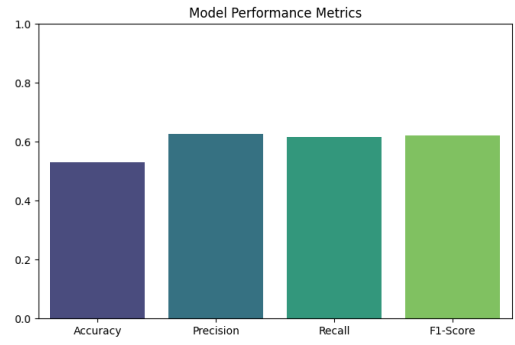


Fig. 4. Model performance matrix bar chart

mandatory because it can have serious consequences in the future. It will be possible to achieve this accuracy in the future through improvements in datasets and other errors in the model.

The results show that the CNN model is used to assist in the diagnosis of early pneumonia using imaging data. The model is biased towards pneumonia because it is given a large number of pneumonia images. The performance of both pneumonia and normal cases is not satisfactory, but the detection ratio of normal people is very low. This model performs well in disease detection, and the model will be practically improved by increasing the number of normal images. In addition to increasing the number of normal images, the quality and quantity of both classes of images should be increased. Then the performance of the model will be significantly improved and will be ready for use in real-life disease detection in the medical field.

Figure 5 shows the model working on the data it was trained on and generalizing on the unseen validation data at each epoch. These are the learning characteristics of the model. The accuracy curve increases and the loss curve decreases, respectively, indicating good generalization.

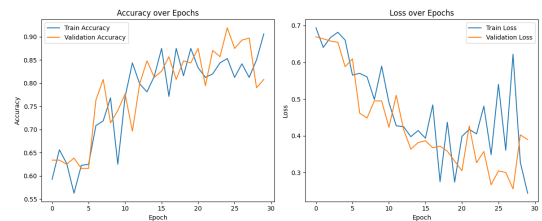


Fig. 5. Showing accuracy and loss curves for the pneumonia detection model

VI. CONCLUSION FUTURE WORK

Abdullahi Umar Ibrahim et al. [13] trained the models based on two binary classes and multiclass. For the two binary classes, COVID-19, non-COVID-19 viral pneumonia and bacterial pneumonia were each classified with healthy CXR images. In this study, the prediction of pneumonia from chest X-rays using CNN model is obtained and the model has good results in predicting pneumonia and normal cases

when given chest X-rays. Using confusion metrics, we get more than 50% accuracy, precision, recall and F1-score for the model. And the model provides very satisfactory results in identifying real patients between pneumonia and normal cases and the ratio is above 60%. Which means the ratio of identifying real patients is quite high.

In the current medical system, chest X-ray is almost mandatory for the diagnosis of pneumonia. In this paper, we have tried to reduce the problem of disease detection using existing data and developed a model. We have developed a model that predicts pneumonia very well. This method is less expensive than the manual method. The detection time is much less than a person can know about his physical condition in a short time. This method requires less skilled person and the overall cost of diagnosis is significantly reduced. Currently, this type of model is easy to adopt even in medical centers and the performance of the model is very effective.

Despite following all the necessary steps, this study faces some limitations. Due to the weakness of the model, it is more inaccurate while predicting normal individuals. Although the quality of the dataset is quite good, the amount of normal chest X-ray images significantly affects the performance of the model. Overall, the performance of the model indicates that in addition to the dataset, data augmentation, balance management, and optimal model selection are also needed. Doing these will definitely improve the performance of the model.

The performance of that model is still not sufficient for real-life use. Insufficient dataset is the main problem of this model. The model can be improved by improving the dataset. The quality and quantity of the dataset is always improving. This improved dataset can be used. New algorithms are also being developed which also improve the performance. Various tools affect the entire process starting from dataset creation, model creation and model management. Those tools can also be integrated. Such improved datasets, algorithms and other tools improve the performance of the model. Therefore, their use will further enrich the performance of the model in the future.

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